

Confidence Intervals for a Population Mean

Let $X_1, \dots, X_n$ be a simple random sample from a population with mean μ and variance σ^2 . Let $\bar{X}$ be the sample mean, and S_n be the sum of sample observation. If n is sufficiently large,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

and

$$S_n \sim N\left(n\mu, n\sigma^2\right)$$

Let $X_1, \dots, X_n$ be a **large** ($n > 30$) random sample from a population with mean μ and standard deviation σ , so that $\bar{X}$ is approximately normal. Then a level $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{X} \pm z_{\frac{\alpha}{2}} \sigma_{\bar{X}}$$

where $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$. When the value of σ is unknown, it can be replaced with the sample standard deviation s .

Small Sample Confidence Intervals for a Population Mean

Let $X_1, \dots, X_n$ be a small ($n < 30$) sample from a *normal* population with mean μ . Then the quantity

$$\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

has a Student's t distribution with $n - 1$ degrees of freedom, denoted t_{n-1} .

When n is large, the distribution of quantity $\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$ is very close to normal, so the normal curve can be used, rather than the Student's t .

Hypothesis Testing

Large-Sample Tests for a Population Mean

Let $X_1, \dots, X_n$ be a **large** ($n > 30$) sample form a population with mean μ and standard deviation σ .

To test a null hypothesis of the form $H_0 : u \leq u_0, H_0 : u \geq \mu_0, H_0 := \mu_0$:

- Compute the z -score:

$$z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

If σ is unknown it may be approximated with s .

- Computer the P -value. The P -value is an area under the normal curve, which depends on the alternate hypothesis as follows:

Alternate Hypothesis	P -Value
$H_1 : \mu > \mu_0$	Area to the right of z
$H_1 : \mu < \mu_0$	Area to the left of z
$H_1 : \mu = \mu_0$	Sum of the areas cut off by z and $-z$

Drawing Conclusions from the Results of Hypothesis Tests

Let α be any value between 0 and 1. Then, if $P \leq \alpha$,

- The result of the test is said to be statistically significant at the $100\alpha\%$ level.
- The null hypothesis is rejected at the $100\alpha\%$ level.
- When reporting the result of the hypothesis test, report the P -value, rather than just comparing it to the 5% or 1%.

Small-Sample Tests for a Population Mean

Let $X_1, \dots, X_n$ be a **small** ($n \leq 30$) random sample from a *normal* population with mean μ (unknown) and a standard deviation σ .

To test a null hypothesis of the form $H_0 : \mu \leq \mu_0, H_0 : \mu \geq \mu_0$, or $H_0 : \mu = \mu_0$:

- Compute the test statistic

$$t^* = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

- Compute the P -value. The P -value is an area under the Student's t curve with $n - 1$ degrees of freedom, which depends on the alternate hypothesis as follows

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- If σ is known, the test statistic is $z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$

Large-Sample Tests for the Difference Between Two Means

Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ be **large** ($n_x > 30$ and $n_y > 30$) independent random samples from populations with mean u_x and u_y and standard deviation σ_x and σ_y , respectively.

The test statistic is as follows:

$$z^* = \frac{(\bar{X} - \bar{Y}) - \delta_0}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}}$$

If σ_X and σ_Y are unknown they may be replaced by s_X and s_Y , respectively

Null Hypothesis	Alternative Hypothesis	p -value
$H_0 : \mu_x - \mu_y \leq \delta_0$	$H_1 : \mu_x - \mu_y > \delta_0$	$P(Z \geq z^*)$
$H_0 : \mu_x - \mu_y \geq \delta_0$	$H_1 : \mu_x - \mu_y < \delta_0$	$P(Z \leq z^*)$
$H_0 : \mu_x - \mu_y = \delta_0$	$H_1 : \mu_x - \mu_y \neq \delta_0$	$2 \times P(Z \geq z^*)$

Small-Sample Tests for the Difference Between Two Means

Population Variances Are Not Equal

Let $X_1, \dots, X_{n,x}$ and $Y_1, \dots, Y_{n,y}$ be samples from *normal* populations with means μ_X and μ_Y and standard deviations σ_X and σ_Y , respectively. Assume the samples are drawn independently of each other.

If σ_x and σ_Y are not known to be equal, then, to test a null hypothesis of the form $H_0 : \mu_X - \mu_Y \leq \Delta_0, H_0 : \mu_X - \mu_Y \geq \Delta_0$, or $H_0 : \mu_X - \mu_Y = \Delta_0$.

- Rounding down to the nearest integer, calculate

$$\nu = \frac{\left[\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}\right]^2}{\frac{\left(\frac{s_X^2}{n_X}\right)^2}{n_X - 1} + \frac{\left(\frac{s_Y^2}{n_Y}\right)^2}{n_Y - 1}}$$

- Compute the test statistic

$$t = \frac{(\bar{X} - \bar{Y}) - \Delta_0}{\sqrt{S_X^2/n_x + S_Y^2/n_y}}$$

- Compute the P -value. The P -value is an area under the Student's t curve with ν degrees of freedom, which depends on the alternate hypothesis as follows:

Alternate Hypothesis	P -value
$H_1 : \mu_X - \mu_Y > \Delta_0$	Area to the right of t
$H_1 : \mu_X - \mu_Y < \Delta_0$	Area to the left of t
$H_1 : \mu_X - \mu_Y \neq \Delta_0$	Sum of the areas in the tails cut off

Population Variances Are Equal

Let $X_1, \dots, X_{n,x}$ and $Y_1, \dots, Y_{n,y}$ be samples from *normal* populations with means μ_X and μ_Y and standard deviations σ_X and σ_Y , respectively. Assume the samples are drawn independently of each other.

If σ_X and σ_Y are known to be qual, then, to test a null hypothesis of the fortm $H_0 : \mu_X - \mu_Y \leq \Delta_0, H_0 : \mu_X - \mu_Y \geq \Delta_0$, or $H_0 : \mu_x - \mu_y = \Delta_0$:

- Compute

$$s_p = \sqrt{\frac{(n_X - 1)s_X^2 + (n_Y - 1)s_Y^2}{n_X + n_Y - 2}}$$

- Compute the test statistic

$$t = \frac{(\bar{X} - \bar{Y}) - \Delta_0}{s_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}}$$

- Compue the P -value. The P -value is an area under the Student's t curve with $n_X + n_Y - 2$ degrees of freedom, which depends on the alternate hypothesis as follows.

Alternate Hypothesis	P -value
$H_1 : \mu_X - \mu_Y > \Delta_0$	Area to the right of t
$H_1 : \mu_X - \mu_Y < \Delta_0$	Area to the left of t
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Correlation vs. Causation

Correlation

A correlation coefficient (denoted r) deasures the strength and direction of a linear relationship between two variables. Let $(x_1, y_1), \dots, (x_n, y_n)$ represent bivariate data, then the correlation coefficient is

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{x_i - \bar{x}}{s_x} \right) = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sqrt{\sum_{i=1}^n x_i^2 - n \bar{x}^2} \sqrt{\sum_{i=1}^n y_i^2 - n \bar{y}^2}} = \frac{SSR}{SST}$$

The Least-Squares Line

For an equation of the form

$$y_1 = \beta_0 + \beta_1 x_i + \epsilon_i$$

y_i is called the **dependent variable**, x_i is called the **indepedent variable**, β is called the **regression coefficients** (the least squares coefficients), and ϵ_i is called the **error**.

Also, r^2 is the **proportion of variance in y explained by regression**.

$$e_1 = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{B}_1 x_i$$

$$\hat{\beta}_1 = \hat{\beta}_1 = \frac{\sum_{i=1}^n y_i - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Uncertainties in the Least-Squares Coefficients

Using some assumptions,

- The quantity $\hat{\beta}$ is *normally distributed* random variables.
- The means of $\hat{\beta}$ is the true values of β .
- The *standard deviations* of β is estimated with

$$s_{\beta_0} = s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

$$s_{\beta_1} = \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

where

$$s = \sqrt{\frac{(1 - r^2) \sum_{i=1}^n (y_i - \bar{y})^2}{n - 2}}$$

is an estimate of the error standard deviation σ .

Confidence Intervals for Coefficients

Under assumptions, the quantities $\frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}}$ and $\frac{\hat{\beta}_1 - \beta_1}{s_{\hat{\beta}_1}}$ have Student's t distributions with $n - 2$ degrees of freedom.

Level $100(1 - \alpha)\%$ confidence intervals for β_0 and β_1 are given by

$$\bar{\beta}_0 \pm t_{n-2} \times s_{\bar{\beta}_0} \quad \bar{\beta}_1 \pm t_{n-2} \times s_{\bar{\beta}_1}$$

Level $100(1 - \alpha)\%$ confidence intervals for the quantity $\beta_0 + \beta_1 x$ is given by

$$\widehat{\beta}_0 + \widehat{\beta}_1 x \pm t_{n-2, \alpha/2} \times s_{\hat{y}}$$

where

$$s_{\hat{y}} = s \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

Checking Assumptions

If the plot of residuals versus fitted values

- Shows no substantial trend or curve, and
- Is **homoscedastic**, that is, the vertical spread does not vary too much along the horizontal length of the plot, except perhaps near the edges,

then it is *likely*, but not certain, that the assumptions of the linear model hold.

However, if the residual plot *does* show a substantial trend or curve, or is **heteroscedastic**, it is certain that the assumptions of the linear plot *do not* hold.

Miscellaneous Notes

- A **test statistic** is a function of the sample data whose value is used to test a hypothesis
- A **p-value** is a measure of the disagreement between a sample and H_0 .
- The smaller the P -value, the more certain we can be that H_0 is false and vice versa.
- For **large samples**, we approximate the population standard deviation σ using the sample standard deviation s .
- The correlation coefficient is called the **sample correlation** (r), and it is an estimate of the population correlation (ρ).
- Some properties of the correlation coefficient (r):

1. $-1 \leq r \leq 1$, r is unitless.
 2. If the points lie exactly on a horizontal or vertical line, the correlation coefficient is undefined, because one of the standard deviations is equal to zero.
 3. Whenever $r \neq 0$, x and y are said to be correlated. If $r = 0$, x and y are said to be uncorrelated.
 4. Correlation coefficient is unaffected by the units in which the measurements are made.
- For **small samples**, s may be far from σ , which invalidates this large-sample method. However, when the population is approximately normal, the Student's t distribution can be used.

- The **pooled** sample variance is

$$s_p^2 = \frac{(n_X - 1)s_X^2 + (n_Y - 1)s_Y^2}{n_X + n_Y - 2}$$

- The correlation coefficient remains unchanged under each of the following operations
 - Multiplying each value of a variable by a positive constant.
 - Adding a constant to each of a variable.
 - Interchanging the values of x and y .
- A **goodness-of-fit statistic** measures how well a model explains a given set of data.
 - SST = total sum of squares = $\sum_{i=1}^n (y_i - \bar{y})^2$
 - SSE = error sum of squares = $\sum_{i=1}^n (y_i - \hat{y})^2$
 - SSR = regression sum of squares = $SST - SSE$
 - The following assumptions are satisfied.
 - The errors $\epsilon_1, \dots, \epsilon_0$ are random and independent. In particular, the magnitude of any error ϵ_i does not influence the value of the next error ϵ_{i+1} .
 - The errors $\epsilon_1, \dots, \epsilon_0$ all have mean 0.
 - The errors $\epsilon_1, \dots, \epsilon_0$ all have the same variance, which we denote by σ^2 .
 - The errors $\epsilon_1, \dots, \epsilon_0$ are normally distributed.
 - Margin of error = $t_{\frac{\alpha}{2}, \sqrt{n}} \times \frac{\sigma}{\sqrt{n}}$
 - The sample variance is calculated

$$\frac{1}{N - 1} \sum_{i=0}^n (x - \bar{x})^2$$